

AI Literature Digest

Technical Guide & Replication Instructions

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This document describes a local web application that automatically retrieves, ranks, tags, and summarises recent PubMed publications on **long-acting antiretrovirals (LA-ARV)** and **broadly neutralising antibodies (bnAbs) for HIV treatment** as an example. It is written so that a colleague can replicate the setup independently.

1. What the App Does

Each time you run the app, it performs the following steps automatically:

1. Searches PubMed across configurable queries, filtered to a chosen date window (default: 14 days).
2. Deduplicates results — a paper returned by multiple queries is counted once.
3. Looks up journal metrics (JIF, SJR, CiteScore) from a built-in database of ~80 journals.
4. Ranks papers by Journal Impact Factor, SCImago Journal Rank, or CiteScore — switchable in the interface.
5. **Tags each paper** automatically using user-defined keyword rules (topic and phase).
6. Sends the ranked paper list to Claude (Anthropic AI) to generate a concise prose digest grouped by theme.
7. Displays the digest with clickable PMID links to PubMed.
8. Sends the digest to your inbox via Gmail SMTP.

 *The app runs entirely on your computer. Data is only sent to PubMed (free, public) and the Anthropic API (pay-per-use — a weekly digest costs a few cents).*

2. Requirements

2.1 Software

- **Node.js** (v18 or later) — nodejs.org
- **A modern browser** (Chrome, Firefox, Edge, Safari)
- **A Gmail account** with 2-Step Verification enabled

2.2 Accounts and Keys

What	Where to get it	Cost
Anthropic API key	console.anthropic.com	~€0.01–0.05 per digest
Gmail App Password	myaccount.google.com/apppasswords	Free

 **Gmail App Password:** a 16-character code generated at myaccount.google.com/apppasswords. It is NOT your regular Gmail password. 2-Step Verification must be enabled first.

3. Installation

3.1 Download

Download `hiv-digest.zip` and unzip it to a folder of your choice, for example `Documents/hiv-digest`.

3.2 Install dependencies

Open a terminal (**Win + R**, then type `cmd`), navigate to the app folder, and run:

```
cd path/to/hiv-digest
npm install
```

This installs three packages: **Express** (local server), **node-fetch** (HTTP), and **Nodemailer** (email). Only needs to be done once.

3.3 Get an Anthropic API key

1. Go to console.anthropic.com and create a free account.
2. Click "API Keys" in the left menu and create a new key.
3. Copy the key — it starts with `sk-ant-`. Paste it into the app each session.

3.4 Create a Gmail App Password

1. Enable 2-Step Verification at myaccount.google.com/security.
2. Go to myaccount.google.com/apppasswords.
3. Choose "Mail" and your device. Click Generate.
4. Copy the 16-character code shown.

4. Running the App

4.1 Start the server

```
npm start
```

You should see: ✓ HIV Digest server running — Open `http://localhost:3000`. Leave this terminal open.

4.2 Open the interface

Go to <http://localhost:3000> in your browser.

4.3 Configure and run

1. Enter your recipient email address.
2. Paste your Anthropic API key into the **Anthropic API key** field.
3. Enter your Gmail address and App Password in the **Gmail credentials** section.
4. Adjust the **search window** (default 14 days) and **max results per query** (default 8) if needed.
5. Edit search queries and tags as needed (see Sections 6 and 7).
6. Click **Run digest**. A progress bar and log panel show each step in real time.
7. Review the digest, then click **Send via Gmail** or **Copy digest**.

4.4 Stop the server

Press `Ctrl+C` in the terminal.

5. How It Works

5.1 Architecture

- **Node.js server** (server.js) — serves the interface, proxies the Anthropic API, looks up journal metrics, and sends email via SMTP.
- **Browser interface** (public/index.html) — all PubMed search logic, paper rendering, sorting, tagging, and filtering runs here.

5.2 PubMed search

The app uses the free NCBI E-utilities API (no key required). Each query is sent as a date-filtered search returning paper IDs, followed by a summary fetch for titles, authors, journals, and publication types. Results are deduplicated by PubMed UID.

5.3 Deduplication

Before adding a paper to the results list, the app checks whether its PubMed UID already exists. If the same paper is returned by two queries, it is only added once. The activity log shows "*X new papers added*" per query so you can see how many duplicates were filtered out.

5.4 Journal ranking

Journal metrics are stored in a local lookup table (journals.js) covering ~80 journals. Three metrics are used:

Metric	Source	What it measures
JIF (Journal Impact Factor)	Clarivate JCR 2023	Avg citations per paper, 2-year window
SJR (SCImago Journal Rank)	SCImago 2023 (open access)	Citation prestige — weights by citing journal quality
CiteScore	Scopus 2023	Avg citations per paper, 4-year window

Tier badges are assigned based on JIF: Q1 (≥ 10), Q2 (5–9.9), Q3 (2–4.9), Q4 (< 2). Sort order can be switched between JIF, SJR, and CiteScore in the interface.

5.5 Tagging

Each paper is tagged automatically using keyword rules defined in the **Tags** configuration panel. A paper is tagged if its title contains any keyword belonging to that tag. Keywords are case-insensitive; trailing `*` enables prefix matching (e.g. `inject*` matches "injectable", "injection").

 *Tagging is based on the paper title only — PubMed summaries do not include abstracts. Most papers are tagged correctly; ambiguous titles may occasionally be missed.*

5.6 AI digest

The ranked, tagged paper list is sent to **Claude Sonnet** via the Anthropic API. The prompt instructs it to write a prose digest grouped by theme, prioritising Q1/Q2 papers, and to add a `[PMID:XXXXXXXX]` marker after each paper discussed. The app converts those markers into clickable links in the browser, and into plain URLs in the email.

5.7 Email

Sending uses **Nodemailer** with Gmail SMTP and a Gmail App Password. The email contains the AI prose summary and a ranked paper list with tier labels and direct PubMed links.

6. Search Queries

Five default queries are pre-loaded. They can be edited, added, or removed in the interface at any time.

- `long-acting antiretroviral HIV clinical trial`
- `broadly neutralising antibodies HIV treatment outcomes`
- `injectable antiretroviral HIV phase 1 phase 2 phase 3`
- `oral long-acting antiretroviral HIV trial`
- `bnAb HIV analytical treatment interruption viral rebound`

PrEP studies are excluded from the AI summary via the prompt instruction (not at the search level, so they may appear in the paper list but will not be discussed in the digest).

7. Tags

7.1 What tags do

Each paper in the results list displays coloured tag chips showing its topic (e.g. bnAb, Injectable LA-ARV) and clinical trial phase (Phase 1/2/3). Keywords used for indexing, search expansion, and filtering should be defined and refined using [the MeSH Browser](#) to ensure consistent terminology and improved retrieval accuracy. The **filter bar** above the paper list lets you click tags to filter — only papers matching **all selected tags** are shown (AND logic). A paper count updates in real time. Click **Show all** to reset filters.

7.2 Default tags

Label	Example keywords
bnAb	bnab, broadly neutralising antibody, vrc01, 3bnc117, 10-1074, n6ls
Injectable LA-ARV	inject, cabotegravir, lenacapavir, rilpivirine, intramuscular
Oral LA-ARV	oral long-acting, oral la-arv, once-weekly, weekly pill
Phase 1	phase 1, phase i trial
Phase 2	phase 2, phase ii trial
Phase 3	phase 3, phase iii trial

7.3 Editing tags

The **Tags** panel is visible in the configuration area, above the Run button. It shows all current tags with their label and keywords.

To edit an existing tag

1. Click the Edit button next to the tag.
2. Update the label and/or keywords in the modal that appears.
3. Click Save.

To add a new tag

1. Click **+ Add tag** in the Tags panel header.
2. Enter a label and a comma-separated list of keywords.
3. Click Save. The new tag is assigned a colour automatically.

To remove a tag

1. Click the Remove button next to the tag.
2. Confirm the deletion.

💡 If you edit or add tags after running a digest, papers are automatically re-tagged, and the filter bar updates immediately. No need to re-run the digest.

7.4 Keyword syntax

Syntax	Example	Matches
Plain word	cabotegravir	Any title containing "cabotegravir"
Phrase	broadly neutralising antibody	Exact phrase in title
Prefix wildcard *	inject*	"injectable", "injection", "injected"...

8. Updating Journal Metrics

The app ships with a built-in database of approximately 80 journals covering HIV, infectious disease, immunology, pharmacology, and major clinical medicine titles. Metrics reflect Clarivate JCR 2023 and SCImago 2023 data. The database can be updated at any time from within the app.

8.1 When to update

Clarivate releases new JIF values each June. SCImago releases updated SJR rankings around the same time. It is good practice to refresh the database once per year after these releases, and to add journals as you encounter them in your results.

8.2 CSV format

Prepare a CSV file with the following columns. A header row is required; column names are case-insensitive. Any metric column can be left empty if the value is not available.

Column	Required	Description
journal	Yes	Full journal name (e.g. The Lancet HIV)
jif	No	Journal Impact Factor (e.g. 17.6)
sjr	No	SCImago Journal Rank score (e.g. 5.2)
citescore	No	Scopus CiteScore (e.g. 20.0)

Example:

```
journal,jif,sjr,citescore
The Lancet HIV,17.6,5.2,20.0
AIDS,4.0,1.6,6.1
Clinical Infectious Diseases,11.8,3.8,15.0
```

SJR data can be downloaded free at scimagojr.com. Search by journal title, select your journals, and export as CSV.

8.3 Importing a CSV

1. Open the app at <http://localhost:3000> (server must be running).
2. Find the Journal metrics panel in the configuration area.
3. Either drag and drop your CSV file onto the drop zone or click the zone to browse for the file. Alternatively, expand or paste CSV directly, paste the text, and click Parse.
4. A preview table appears showing all rows with their parsed values and a status badge — New for journals not currently in the database, Update for those that will be overwritten. Review this before proceeding.
5. Click Import to apply. A confirmation message shows how many journals were added and updated.

8.4 Behaviour after import

Changes take effect immediately. If a digest has already been run in the current session, papers are re-ranked on the spot using the updated metrics — no need to re-run the digest. The tier badge for each journal (Q1–Q4) is recalculated automatically based on the new JIF value using the thresholds: Q1 ≥ 10 , Q2 5–9.9, Q3 2–4.9, Q4 < 2 .

8.5 Persistence

Imported data is held in memory for the current server session. When the server is restarted, it reloads from the original journals.js file. To make changes permanent, open journals.js in a text editor and add or update entries following the existing format — each entry requires a lowercase journal name, jif, sjr, citeScore, sjrQ, and tier.

9. Customising for Another Field

The app is not HIV-specific at the code level. To adapt it:

1. **Change the search queries** in the interface.
2. **Update the journal table** — open `journals.js` in a text editor and add journals with their JIF, SJR, and CiteScore. SJR data is freely available at scimagojr.com.
3. **Update the tags** — use the Tags panel in the interface to redefine labels and keywords for your field.
4. **Update the Claude prompt** — in `public/index.html`, search for "You are a clinical research assistant" and update the description and grouping headings.

10. File Structure

File	Purpose
server.js	Node.js server: Anthropic API proxy, journal metrics, email (SMTP)
journals.js	Journal metrics lookup table (JIF, SJR, CiteScore, tier)
public/index.html	Browser interface: all search, render, tag, filter, and digest logic
package.json	Node.js project config and dependencies
README.md	Quick-start instructions

11. Troubleshooting

No papers found

Widen the search window to 30 or 60 days using the slider. You can also simplify a query (e.g. `long-acting HIV`) to broaden it.

"Failed to fetch" error

The server is not running. Open a terminal in the app folder, run `npm start`, then reload the page.

Email not sending

Check that: (1) your Gmail account has 2-Step Verification enabled, (2) you are using a Gmail App Password (not your regular password), (3) the App Password was entered without spaces.

Journal shown as "-" (not in database)

Open `journals.js`, copy an existing entry, and add the journal with its metrics. SJR data is available free at scimagojr.com.

Claude API error

Check your API key and account credits at console.anthropic.com. A digest costs \$0.01–0.05.

Tags not matching expected papers

Open the Tags panel and click Edit on the relevant tag. Check whether the keyword matches the exact wording in paper titles — PubMed titles can vary. Add alternative phrasings as extra keywords (comma-separated).

12. Modifying the App with Claude

All the files in this app are plain text and can be uploaded directly into a **Claude.ai** conversation for editing. You do not need any coding experience to make changes — describe what you want in plain language and Claude will modify the file and return it ready to use.

12.1 Which files can be uploaded

File	What you can ask Claude to change
public/index.html	Interface layout, tag defaults, digest prompt, filter behaviour, sort options, new features
server.js	Email settings, new API endpoints, server behaviour
journals.js	Add journals, update metrics, change tier thresholds
HIV_Digest_Guide.docx	Update documentation, add sections, change wording

12.2 How to do it

1. Go to claude.ai and start a new conversation.
2. Upload the file you want to modify using the attachment button.
3. Describe the change you want in plain language. Be specific about what to change and what to leave alone.
4. Claude will return the modified file. Download it and replace the original in your app folder.
5. Restart the server (`npm start`) to pick up changes to `server.js` or `journals.js`. Changes to `index.html` take effect on the next browser reload.

12.3 Example requests

- *"In index.html, add a new sort option that sorts papers by publication date within each tier group."*

- *"In index.html, change the digest word limit from 700 to 1000 words and ask Claude to include a one-line plain-language summary for each paper."*
- *"In journals.js, add these five journals with the following metrics..."*
- *"In index.html, only change the email section to also send a plain-text version with no formatting."*

 **Tip:** always keep a backup copy of the original file before uploading it for edits. The zip file you downloaded is a good baseline to preserve.

12.4 Practical tip for large files

`index.html` is approximately 1,400 lines. Claude handles files of this size well, but if you are making several rounds of changes it helps to be explicit: *"Only modify the tag manager section — do not change anything else."* This reduces the risk of unintended edits elsewhere in the file.

13. Quick-Start Checklist

Use this checklist the first time you set up the app.

- **Download:** unzip hiv-digest.zip to a folder
- **Install:** open terminal in that folder → `npm install`
- **Anthropic key:** create at console.anthropic.com → copy key starting with sk-ant-
- **Gmail App Password:** enable 2-Step Verification → generate at myaccount.google.com/apppasswords
- **Run:** `npm start` → open <http://localhost:3000>
- **Configure:** enter email, API key, Gmail credentials
- **Tags:** review default tags in the Tags panel, edit or add as needed
- **Run digest:** click Run digest → review → filter by tag → Send via Gmail

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